MINISTRY OF EDUCATION AND TRAINING MINISTRY OF HEALTH

HANOI MEDICAL UNIVERSITY

PHAM HOANG ANH

PREVENTIVE MEDICINE

HPV VACCINATION UPTAKE AND SOCIOECONOMIC INEQUALITY AMONG VIETNAMESE WOMEN AGED 15–29 YEARS:

A SECONDARY DATA ANALYSIS OF THE 2020–2021 MICS 6 SURVEY

MEDICAL DOCTOR GRADUATION THESIS

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1. Assoc. Prof. Pham Bich Diep **2. Dr. Pham Thanh Tung**

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Above all, I owe an immeasurable debt to my family for their unconditional love, patience, and sacrifice. You are my greatest motivation.

Hanoi, 2026

PHAM HOANG ANH

DECLARATION

I hereby solemnly declare that this thesis, entitled “HPV Vaccination Uptake and Socioeconomic Inequality Among Vietnamese Women Aged 15–29 Years: A Secondary Data Analysis of the 2020–2021 MICS 6 Survey,” is my original work, conducted under the supervision of Assoc. Prof. Pham Bich Diep and Dr. Pham Thanh Tung. The data, statistical analyses, and findings presented herein are truthful and have not been previously submitted for any other degree or qualification at this or any other institution.

All referenced sources, including published literature, datasets, and methodological frameworks, are duly cited in accordance with the academic citation standards prescribed by Hanoi Medical University. The Viet Nam MICS6 dataset used in this study was accessed in compliance with the official data use agreement and ethical protocols.

I take full responsibility for the integrity of the research presented in this thesis.

Hanoi, 2026

Student

PHAM HOANG ANH

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ABSTRACT

Background. Human papillomavirus (HPV) vaccination is a highly effective primary prevention strategy for cervical cancer, yet vaccine awareness and uptake remain unequally distributed in many low- and middle-income settings. National evidence on socioeconomic inequality in HPV vaccine awareness and vaccination among young Vietnamese women is limited.

Objectives. This thesis aimed to estimate the prevalence of HPV vaccine awareness and HPV vaccination among Vietnamese women aged 15–29 years, assess associated sociodemographic factors, quantify wealth-related inequality using concentration indices, and decompose the observed contributors to inequality.

Methods. A cross-sectional secondary analysis was conducted using the Viet Nam Multiple Indicator Cluster Survey 2020–2021. The analytic sample included 4,102 women aged 15–29 years with valid survey design information. All descriptive and bivariate analyses accounted for sampling weights, stratification, and clustering. Bivariate p-values were obtained from design-based Pearson chi-squared tests with survey correction. Adjusted prevalence ratios (aPRs) were estimated using survey-weighted Poisson regression. Wealth-related inequality was measured using the standard concentration index and the Erreygers-corrected concentration index; decomposition used probit average marginal effects.

Results. Overall, 62.4% of women had heard of HPV vaccination (95% CI: 59.8–64.9), while 7.5% had received HPV vaccine (95% CI: 6.2–8.7). Among women who were aware of HPV vaccination, only 12.0% were vaccinated (95% CI: 10.1–13.9), leaving an awareness–vaccination gap of 88.0% (95% CI: 86.1–89.9). Both awareness and vaccination were concentrated among wealthier women. The Erreygers concentration index was 0.371 (95% CI: 0.328–0.415) for awareness and 0.116 (95% CI: 0.089–0.144) for vaccination. The gap was concentrated among poorer aware women (Erreygers CI: -0.123; 95% CI: -0.164 to -0.082). In adjusted models, the poorest quintile had substantially lower awareness (aPR 0.576; 95% CI: 0.480–0.691) and vaccination (aPR 0.118; 95% CI: 0.039–0.358) than the richest quintile, and a higher prevalence of remaining unvaccinated despite awareness (aPR 1.137; 95% CI: 1.055–1.225).

Conclusion. HPV vaccine awareness is moderately common among young Vietnamese women, but vaccination uptake remains low. Socioeconomic position, education, residence, media exposure, ethnicity, and gender-equitable attitudes were associated with different stages of the awareness-to-vaccination pathway. These estimates provide a pre-program baseline for equity monitoring as HPV vaccination is incorporated into public immunization financing from 2026. Equity-oriented HPV vaccination policy should combine public financing with targeted outreach for poorer, rural, less educated, and ethnic minority women.

Keywords: HPV vaccination; cervical cancer prevention; socioeconomic inequality; concentration index; Erreygers decomposition; MICS; Viet Nam.

LIST OF ABBREVIATIONS

Abbreviation	Full term
aPR	Adjusted Prevalence Ratio
CI	Confidence Interval; Concentration Index
CIN	Cervical Intraepithelial Neoplasia
ECI	Erreygers-Corrected Concentration Index
EPI	Expanded Programme on Immunization
GSO	General Statistics Office of Viet Nam
HPV	Human Papillomavirus
ICT	Information and Communication Technology
IRB	Institutional Review Board
LMIC	Low- and Middle-Income Country
MICS	Multiple Indicator Cluster Survey
OR	Odds Ratio
PR	Prevalence Ratio
PSU	Primary Sampling Unit
SES	Socioeconomic Status
UNICEF	United Nations Children’s Fund
WHO	World Health Organization

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INTRODUCTION

Cervical cancer remains one of the most preventable causes of cancer morbidity and mortality among women. Persistent infection with high-risk human papillomavirus (HPV) types is the necessary cause of nearly all cervical cancer, and prophylactic HPV vaccination can prevent the majority of HPV-related cervical precancer and cancer when delivered before or around the period of sexual debut [1, 2]. The global strategy for cervical cancer elimination therefore places HPV vaccination alongside screening and treatment as a central public health intervention [3].

Despite this strong biological and programmatic rationale, HPV vaccination is not only a biomedical issue. In countries where vaccination depends partly on private payment, geographic availability, and individual information-seeking, uptake can become strongly patterned by household wealth, education, residence, ethnicity, and media access. These inequalities matter because they can reproduce the same social gradient already seen in cervical cancer screening and treatment access: women with fewer resources may be less likely to receive both primary prevention and early detection.

Viet Nam provides an important setting for examining this issue. The Viet Nam Multiple Indicator Cluster Survey 2020–2021 (MICS6) collected nationally representative information on women, households, media exposure, reproductive health, and HPV vaccination awareness and uptake [4]. The survey predates the planned inclusion of cervical cancer prevention vaccine in the Expanded Programme on Immunization from 2026 under Resolution 104/NQ-CP [5, 6]. Among young women aged 15–29 years, the dataset therefore permits analysis of both early adult vaccination experience and a pre-program equity baseline before public immunization financing is expanded.

This thesis focuses on three policy-relevant outcomes: having heard of HPV vaccination, having received HPV vaccine, and remaining unvaccinated despite awareness. The last outcome is especially important because it separates a purely informational gap from a practical access gap. A woman may know about the vaccine but remain unable to receive it because of cost, distance, limited service availability, family or social constraints, or low perceived priority.

The specific objectives were:

1. To estimate the survey-weighted prevalence of HPV vaccine awareness, HPV vaccination, and the awareness–vaccination gap among Vietnamese women aged 15–29 years.

2. To describe how these outcomes vary by sociodemographic and access-related characteristics, using survey-corrected bivariate tests.

3. To estimate adjusted associations between explanatory variables and each outcome using survey-weighted Poisson regression.

4. To quantify socioeconomic inequality using standard and Erreygers-corrected concentration indices.

5. To decompose the Erreygers concentration index to identify the main contributors to observed inequality.

CHAPTER 1
LITERATURE REVIEW

1.1 Human papillomavirus and cervical cancer

Cervical cancer is a major global public health problem and remains disproportionately concentrated in settings where prevention services are less accessible. WHO estimated approximately 660,000 new cervical cancer cases and 350,000 deaths globally in 2022, with the greatest burden in low- and middle-income countries [7]. Broader global analyses show that unequal access to screening, diagnosis, treatment, and vaccination underlies much of the international variation in cervical cancer incidence and mortality [8]. In Viet Nam, the ICO/IARC HPV Information Centre estimated that 4,132 women are diagnosed with cervical cancer and 2,223 women die from the disease each year; cervical cancer ranks fifth among cancers in women aged 15–44 years [9].

Human papillomaviruses are double-stranded DNA viruses with more than 200 identified genotypes. Approximately 40 types infect the anogenital tract, and a smaller group is considered high-risk because of oncogenic potential [10]. Persistent infection with high-risk HPV types, particularly HPV 16 and HPV 18, can lead to cervical intraepithelial neoplasia and invasive cervical cancer through a multi-step process involving viral persistence, cellular transformation, and accumulation of precancerous lesions [1, 11].

Most HPV infections clear spontaneously, but persistent high-risk infection can progress over years. This long natural history creates two prevention opportunities: primary prevention through vaccination before infection, and secondary prevention through screening and treatment of precancerous lesions. The present thesis focuses on the primary prevention pathway.

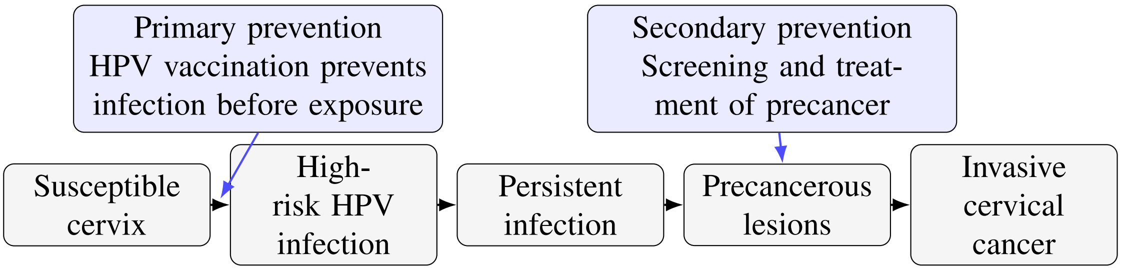

Figure 1.1. Natural history of HPV infection and cervical cancer development. Author-drawn schematic based on established HPV natural history: vaccination prevents infection, while screening detects precancerous lesions for treatment.

1.2 HPV vaccines and prevention

Prophylactic HPV vaccines target the viral types responsible for most cervical cancers and several other HPV-related diseases. HPV 16 and HPV 18 are the dominant oncogenic types globally and account for most cervical cancers; in Viet Nam, 82.8% of invasive cervical cancers are attributed to HPV 16 or 18 [9]. Currently available prophylactic products include bivalent, quadrivalent, and nonavalent vaccines, which differ in the HPV types covered but share the goal of preventing infection before exposure [12]. Randomized trials and population-level studies have shown high efficacy and effectiveness against vaccine-type HPV infection and cervical precancer, especially when vaccination occurs before HPV exposure [13, 14]. WHO's 2022 position paper recommends HPV vaccination as a core intervention for cervical cancer prevention and supports simplified schedules that can improve feasibility in national programs [2]. In 2024, WHO updated product-choice guidance and confirmed an additional prequalified product for single-dose use, reinforcing the programmatic relevance of single-dose schedules for supply, cost, and equity [12,15].

Population impact depends on coverage and equity. Countries with publicly financed and school-based delivery have often achieved high coverage, while countries relying on private-sector delivery or fragmented financing tend to show lower and more unequal uptake [16, 17]. Therefore, measuring vaccine coverage alone is insufficient; the distribution of coverage across socioeconomic groups is also essential.

1.3 HPV vaccination in Viet Nam

Viet Nam has made progress in reproductive, maternal, and child health, but HPV vaccination had not historically reached all eligible girls and young women through a universal nationwide program at the time of MICS6. The National Action Plan on Prevention and Control of Cervical Cancer for 2016–2025 set a target that at least 25% of girls and women would be vaccinated against HPV by 2025, but HPV vaccine was still not delivered through a universal national program during MICS6 data collection [4, 18]. As a result, vaccination experience among young women may reflect household ability to pay, proximity to services, awareness, and local availability. The 2020–2021 MICS6 survey offers nationally representative data that can be used to assess this pattern among women aged 15–29 years before planned public financing of HPV vaccination. In 2022, the Government issued Resolution 104/NQ-CP on increasing the number of vaccines in the Expanded Programme on Immunization, including cervical cancer prevention vaccine from 2026; Government communication at that time described free cervical cancer vaccination for girls from 2026 [5, 6].

The age group in this thesis was chosen for three reasons. First, WHO identifies girls aged 9–14 years as the priority group for HPV vaccination before sexual debut [2, 7]; women aged 15–29 years therefore represent older adolescents and young adults for whom catch-up needs and missed opportunities can be evaluated. Second, it captures early adult women for whom awareness and uptake may be influenced by education, media exposure, marital status, sexual experience, and private-market access before public financing. Third, the group remains large enough in MICS6 to support survey-weighted modeling and inequality measurement. The MICS indicator records whether the respondent has received HPV vaccine, but it does not capture dose number, vaccine product, age at vaccination, price paid, or provider type; therefore, the outcome should be interpreted as self-reported receipt of any HPV vaccine dose rather than confirmed completion of a recommended schedule.

1.4 Social determinants of HPV vaccination

HPV vaccination behavior is shaped by social determinants operating at multiple levels. The social determinants of health framework emphasizes that health behaviors and service use are structured by economic resources, education, gender norms, ethnicity, and place of residence [19]. Andersen's behavioral model of health service use further distinguishes predisposing factors, enabling resources, and perceived or evaluated need [20].

In HPV vaccination, wealth and education may influence ability to pay, health literacy, navigation of services, and trust in preventive care. Urban residence may increase proximity to vaccination providers and exposure to health promotion. Ethnic minority status may reflect language barriers, geographic marginalization, and unequal access to services. Media and ICT exposure can increase awareness, while gender-equitable attitudes may reflect broader empowerment and receptivity to preventive health messages [21–23].

1.5 Measuring socioeconomic inequality

Simple group comparisons describe differences between categories, but they do not summarize the full socioeconomic gradient. The concentration index is widely used to quantify wealth-related inequality in health. It is defined as twice the area between the concentration curve and the line of equality and can also be expressed as a covariance between the health outcome and the fractional rank in the socioeconomic distribution [24, 25]:

$$CI = \frac{2}{\mu}\mathrm{cov}(y_i, r_i), \tag{1.1}$$

where y_i is the health variable, μ is its mean, and r_i is the fractional socioeconomic rank. A positive concentration index indicates that the outcome is concentrated among wealthier individuals; a negative value indicates concentration among poorer individuals.

For bounded outcomes such as binary vaccination indicators, the standard concentration index has limitations because its bounds depend on the outcome mean. Erreygers proposed a correction that is appropriate for bounded variables and more comparable across outcomes with different prevalence [26]. For a binary variable bounded between 0 and 1, the Erreygers-corrected concentration index can be written as:

$$E(y) = 4\mu CI. \tag{1.2}$$

The Erreygers index can be decomposed to estimate the contribution of determinants:

$$E(y) = 4\sum_{k} \beta_k \bar{x}_k CI_k + \varepsilon, \tag{1.3}$$

where β_k is the marginal effect of determinant x_k, $\bar{x}_k$ is its mean, CI_k is its concentration index, and ε is the residual component [26, 27]. This decomposition helps translate inequality measurement into policy interpretation by ranking the factors that contribute most to the observed gradient.

1.6 Conceptual framework

This thesis conceptualizes HPV vaccination as a two-stage prevention pathway. The first stage is awareness: whether a woman has heard of HPV vaccination. The second stage is uptake: whether awareness is translated into actual vaccination. A third outcome, the awareness–vaccination gap, captures women who have passed the awareness stage but remain unvaccinated.

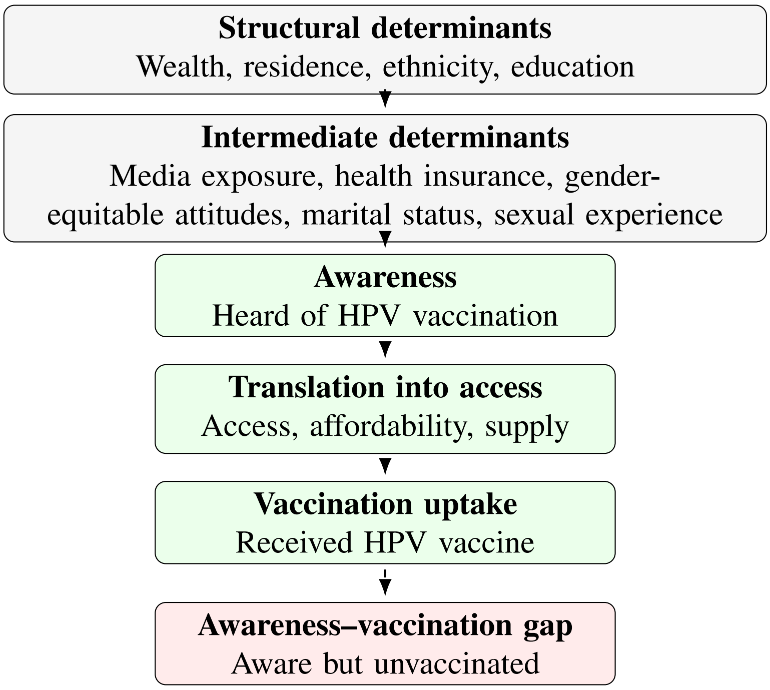

Figure 1.2. Conceptual framework for determinants of HPV vaccine awareness, uptake, and the awareness–vaccination gap among women aged 15–29 years.

The framework treats wealth, education, residence, and ethnicity as structural determinants. Media exposure, ICT exposure, health insurance, gender-equitable attitudes, marital status, and sexual experience are treated as intermediary or individual-level determinants. These factors may influence awareness and uptake differently. For example, media exposure may be more strongly related to awareness, while wealth may become more important when a woman must access and pay for vaccination.

CHAPTER 2
MATERIALS AND METHODS

2.1 Study design and data source

This was a cross-sectional secondary analysis of the Viet Nam Multiple Indicator Cluster Survey 2020–2021 (MICS6), conducted by the General Statistics Office of Viet Nam with technical support from UNICEF. MICS6 is a nationally representative household survey using a stratified, two-stage cluster sampling design based on the 2019 Viet Nam Population and Housing Census. The sample was stratified by urban and rural areas, socioeconomic region, major cities, and ethnic-minority composition of enumeration areas. In the first stage, 700 enumeration areas were selected with probability proportional to size; in the second stage, 20 households were systematically selected within each enumeration area, yielding 14,000 selected households. The women's questionnaire was administered to eligible women aged 15–49 years in selected households; the full women's module interviewed 10,770 women before this thesis restricted the sample to women aged 15–29 years [4].

The analysis used the women's individual dataset in international Stata format. All analyses accounted for the complex survey design, including sampling weights, primary sampling units, and strata.

2.2 Study population

The target population for this thesis was Vietnamese women aged 15–29 years. The analytic sample was defined by the following criteria:

- Women in MICS age groups WAGE 1–3, corresponding to ages 15–19, 20–24, and 25–29 years.
- Non-missing and positive women's sampling weight (`wmweight`).

- Non-missing primary sampling unit (PSU) and stratum (stratum).

After restricting the dataset to women aged 15–29 years and excluding records without valid survey design information, the final analytic sample included **4,102** women. No additional sample size calculation was performed because the study used all eligible observations in the nationally representative dataset.

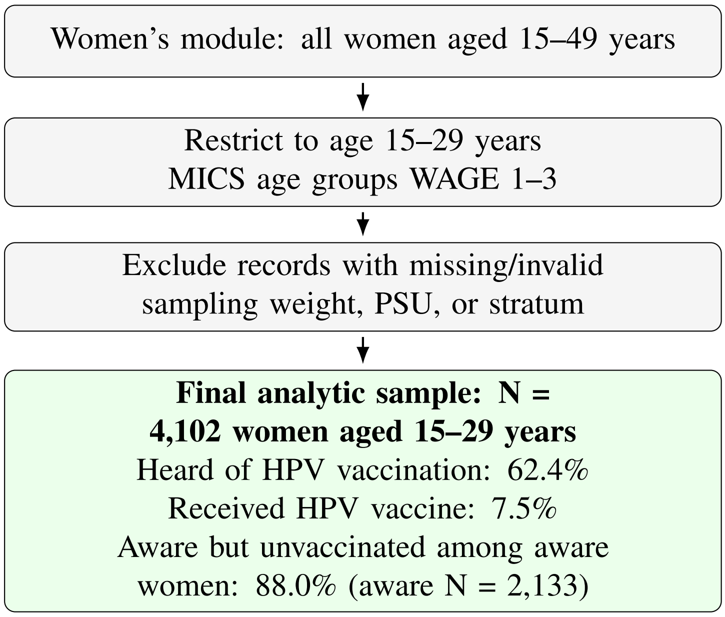

Figure 2.1. Study design and participant selection. The analytic sample included 4,102 women aged 15–29 years from Viet Nam MICS6 with valid survey design variables.

2.3 Outcome variables

Three primary binary outcomes and one sensitivity outcome were constructed:

1. **HPV vaccine awareness** (ccp5_bin): coded 1 if the respondent reported having heard of HPV vaccination and 0 otherwise.

2. **HPV vaccination status in the full analytic sample** (ccp6_bin): coded 1 if the respondent reported having received HPV vaccine and 0 otherwise. Women who had not heard of HPV vaccination were coded as unvaccinated, consistent with the MICS skip pattern. This outcome should be interpreted as reported

vaccination in the full population, under the assumption that women who had never heard of HPV vaccination had not been vaccinated.

3. **Awareness–vaccination gap** (`gap`): defined only among women who had heard of HPV vaccination. It was coded 1 for aware but unvaccinated and 0 for aware and vaccinated.

4. **HPV vaccination among aware women** (`ccp6_aware`): coded 1 for vaccinated and 0 for unvaccinated among women who had heard of HPV vaccination. This was used as a sensitivity analysis to separate the question of awareness from the question of uptake conditional on awareness.

2.4 Explanatory variables and confounder measures

Variables were selected based on the conceptual framework, prior literature, and availability in MICS6. Table 2.1 lists all explanatory variables used in descriptive and bivariate analyses. ICT exposure was included descriptively and bivariately, but it was excluded from adjusted regression and decomposition because its weighted prevalence was 97.9%, leaving sparse contrast and a high risk of unstable estimates.

2.5 Statistical analysis

All statistical analyses were performed in StataNow 19.5 SE. The survey design was declared as:

```
svyset PSU [pweight=wmweight], strata(stratum)
vce(linearized) singleunit(centered)
```

2.5.1 Descriptive analysis

Survey-weighted percentages and 95% confidence intervals were estimated for all outcomes and explanatory variables. These estimates used the `svy` prefix and therefore accounted for weights, clustering, and stratification.

Table 2.1. Explanatory variables and operational definitions

Variable	Categories	Operational definition
Age group	15–19, 20–24, 25–29	MICS age group variable WAGE
Residence	Urban, rural	Household area variable
Ethnicity	Kinh/Hoa, ethnic minority	MICS two-category ethnicity variable
Marital status	Currently married/in union, formerly married/in union, never married/in union	MICS marital status variable
Education	Primary or less, lower secondary, upper secondary or higher	Collapsed from MICS education level
Wealth quintile	Q1 poorest to Q5 richest	MICS household wealth quintile
Health insurance	Insured, uninsured	Recode of health insurance variable
Mass media exposure	No, yes	Any exposure to newspaper, radio, or television
ICT exposure	No, yes	Any reported computer, internet, or mobile phone exposure/use item in the media module
Gender-equitable attitude	No, yes	Yes if the respondent rejected all five wife-beating justifications
Sexual experience	No, yes	Ever had sexual intercourse

2.5.2 Bivariate analysis

For each explanatory variable, outcome prevalence was estimated using survey-weighted proportions. Bivariate p-values were obtained from design-based Pearson chi-squared tests from `svy: tabulate, pearson`. For the gap outcome, all bivariate analyses used `svy, subpop(ccp5_bin)` so that inference was restricted to women who had heard of HPV vaccination while preserving the original survey design.

2.5.3 Adjusted regression models

Adjusted prevalence ratios were estimated using survey-weighted Poisson regression with a log link. This approach was chosen because odds ratios can overstate relative associations when outcomes are common, particularly for HPV vaccine awareness [28,29]. The `svy` prefix used Taylor linearized survey variance estimation, accounting for stratification, clustering, and sampling weights; this provides design-based standard errors for the modified Poisson models. Models included age group, residence, ethnicity, marital status, education, wealth quintile, insurance, mass media exposure, gender-equitable attitude, and sexual experience. ICT exposure was excluded from adjusted models because of near-universal prevalence.

For awareness and vaccination, models were fitted in the full analytic sample. For the awareness–vaccination gap and vaccination-among-aware sensitivity outcome, models used the aware subpopulation while preserving the survey design. Logistic regression models estimating odds ratios were conducted as sensitivity analyses and reported in the appendix. Adjusted predicted probabilities by wealth quintile were also estimated from survey-weighted logistic models to present absolute differences alongside prevalence ratios.

2.5.4 Concentration index and decomposition

Socioeconomic ranking used the continuous household wealth score (`ses`). Wealth quintiles were used for grouped descriptive analysis and as dummy variables in adjusted and decomposition models.

The standard concentration index was estimated for awareness, vaccination, and the awareness–vaccination gap. Because all outcomes were binary, the Erreygers-corrected concentration index was also estimated. Positive values indicate pro-rich concentration; negative values indicate pro-poor concentration.

Decomposition of the Erreygers index used a probit model to obtain average marginal effects for explanatory variables. Categorical determinants were entered as dummy variables, with reference categories matching the adjusted regression models: urban residence, age 25–29 years, currently married/in union, upper secondary or higher education, insured, Kinh/Hoa ethnicity, richest wealth quintile, no mass media exposure, no gender-equitable attitude, and no sexual experience. For each determinant dummy, its own survey-weighted standard concentration index was estimated using the continuous wealth score as the rank variable. The contribution of each determinant was calculated as:

$$\text{Contribution}_k = 4 \times AME_k \times \bar{x}_k \times CI_k, \tag{2.1}$$

where AME_k is the average marginal effect, $\bar{x}_k$ is the weighted mean of the determinant, and CI_k is the determinant's standard concentration index. Percentage contribution was calculated by dividing each contribution by the total Erreygers index. Bootstrap confidence intervals for decomposition contributions were not estimated; therefore, decomposition results are presented as explanatory accounting of the observed inequality pattern and not as causal attribution.

2.6 Ethical considerations

This study used de-identified secondary data from MICS6. The original survey obtained ethical approval and respondent consent according to UNICEF and national survey procedures [4]. No direct contact with human participants occurred in this thesis.

CHAPTER 3
RESULTS

3.1 Analytic sample and outcome prevalence

The final analytic sample included 4,102 women aged 15–29 years. All estimates in this chapter are survey-weighted unless otherwise stated. The overall prevalence of HPV vaccine awareness was 62.4% (95% CI: 59.8–64.9), while HPV vaccination prevalence was 7.5% (95% CI: 6.2–8.7). Among 2,133 women who had heard of HPV vaccination, only 12.0% were vaccinated (95% CI: 10.1–13.9), leaving an awareness–vaccination gap of 88.0% (95% CI: 86.1–89.9).

Table 3.1. Survey-weighted characteristics of women aged 15–29 years, Viet Nam MICS6

Characteristic	Category	% (95% CI)
Age group	15–19	30.4 (28.3–32.4)
	20–24	29.7 (27.4–31.9)
	25–29	39.9 (37.7–42.2)
Residence	Urban	37.4 (32.6–42.1)
	Rural	62.6 (57.9–67.4)
Ethnicity	Kinh/Hoa	85.0 (82.9–87.0)
	Ethnic minority	15.0 (13.0–17.1)
Marital status	Currently married/union	46.4 (43.9–48.9)
	Formerly married/union	3.3 (2.5–4.1)
	Never married/union	50.3 (47.8–52.8)
Education	Primary or less	5.2 (4.2–6.2)
	Lower secondary	22.1 (19.9–24.3)
	Upper secondary+	72.7 (70.2–75.2)
Wealth quintile	Q1 poorest	19.2 (16.9–21.5)
	Q2	21.2 (18.9–23.5)
	Q3	21.9 (19.6–24.2)
	Q4	20.1 (18.0–22.2)
	Q5 richest	17.6 (15.4–19.8)
Health insurance	Insured	86.3 (84.2–88.4)
	Uninsured	13.7 (11.6–15.8)
Mass media exposure	No	9.1 (7.6–10.6)
	Yes	90.9 (89.4–92.4)
ICT exposure	No	2.1 (1.6–2.6)
	Yes	97.9 (97.4–98.4)
Rejects all wife-beating justifications	No	9.8 (8.2–11.4)
	Yes	90.2 (88.6–91.8)
Ever had sexual intercourse	No	46.9 (44.4–49.4)
	Yes	53.1 (50.6–55.6)
Heard about HPV vaccination		62.4 (59.8–64.9)
Received HPV vaccine		7.5 (6.2–8.7)
Aware but unvaccinated, among aware women		88.0 (86.1–89.9)

3.2 Bivariate patterns in awareness and vaccination

Table 3.2 shows outcome prevalence by sociodemographic characteristics. The p-values are from design-based Pearson chi-squared tests using `svy: tabulate, pearson`; percentages and 95% confidence intervals are survey-weighted proportions.

Table 3.2. HPV vaccine awareness and vaccination by explanatory variables

Variable	Category	Awareness % (95% CI)	p	Vaccination % (95% CI)	p
Age	15–19	48.3 (44.2–52.4)	< 0.001	5.6 (3.9–8.0)	0.015
	20–24	66.2 (61.9–70.1)		10.0 (7.7–12.8)	
	25–29	70.3 (67.0–73.4)		7.0 (5.5–8.9)	
Residence	Urban	73.1 (69.0–76.8)	< 0.001	11.4 (9.1–14.1)	< 0.001
	Rural	56.0 (52.6–59.3)		5.1 (3.9–6.7)	
Ethnicity	Kinh/Hoa	67.0 (64.2–69.8)	< 0.001	8.5 (7.2–10.1)	< 0.001
	Ethnic minority	36.1 (31.7–40.6)		1.6 (0.8–3.4)	
Health insurance	Insured	63.0 (60.3–65.6)	0.233	8.0 (6.7–9.5)	0.016
	Uninsured	58.7 (51.7–65.4)		4.0 (2.3–7.0)	
Marital status	Current	63.6 (60.4–66.7)	0.286	5.1 (3.9–6.5)	< 0.001
	Former	54.1 (41.0–66.7)		7.3 (3.4–15.0)	
	Never	61.8 (58.3–65.2)		9.7 (7.7–12.1)	
Education	Primary or less	22.9 (16.0–31.6)	< 0.001	0.8 (0.1–4.0)	< 0.001
	Lower secondary	49.2 (44.3–54.1)		2.4 (1.3–4.3)	
	Upper secondary+	69.2 (66.4–71.9)		9.5 (8.0–11.3)	
Wealth	Q1 poorest	33.3 (29.0–37.9)	< 0.001	0.7 (0.3–1.7)	< 0.001
	Q2	58.5 (53.6–63.3)		6.0 (4.1–8.9)	
	Q3	64.4 (59.3–69.1)		5.9 (4.1–8.5)	
	Q4	73.5 (68.8–77.8)		9.2 (6.7–12.4)	
	Q5 richest	83.6 (79.5–87.0)		16.6 (12.7–21.3)	
Mass media	No	39.1 (32.8–45.7)	< 0.001	2.6 (1.0–6.9)	0.017
	Yes	64.7 (62.1–67.3)		8.0 (6.7–9.4)	
ICT exposure	No	23.4 (15.1–34.5)	< 0.001	2.7 (0.4–16.6)	0.257
	Yes	63.2 (60.6–65.8)		7.6 (6.4–9.0)	
Gender-equitable attitude	No	45.2 (38.5–52.0)	< 0.001	3.2 (1.8–5.9)	0.003
	Yes	65.2 (62.4–67.8)		7.9 (6.7–9.4)	
Ever had sex	No	61.1 (57.5–64.6)	0.271	9.8 (7.8–12.3)	< 0.001
	Yes	63.5 (60.3–66.5)		5.4 (4.3–6.8)	

Note: All estimates account for sampling weights, strata, and primary sampling units. P-values are design-based Pearson chi-squared tests with survey correction.

The strongest unadjusted gradients were observed for wealth, education, residence, and ethnicity. Awareness increased from 33.3% in the poorest quintile to 83.6% in the richest quintile. Vaccination increased from 0.7% in the poorest quintile to 16.6% in the richest quintile. Ethnic minority women had markedly lower awareness (36.1%) and vaccination (1.6%) than Kinh/Hoa women.

3.3 Awareness–vaccination gap

Among women who had heard of HPV vaccination, the gap was highest among poorer and more disadvantaged groups. Table 3.3 presents survey-weighted gap prevalence. P-values are design-based Pearson tests using the aware subpopulation.

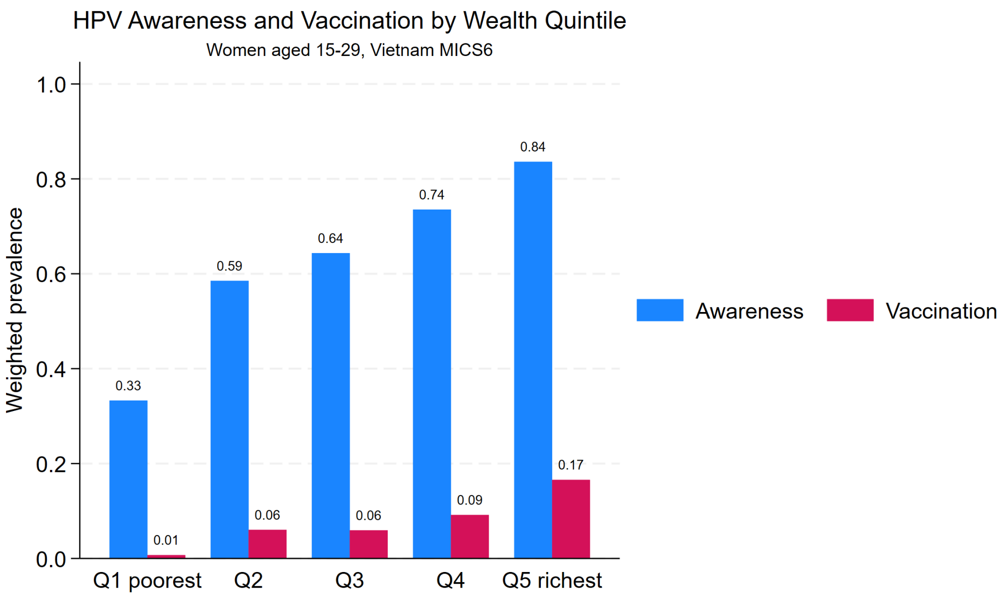

Figure 3.1. HPV vaccine awareness and vaccination prevalence by wealth quintile.

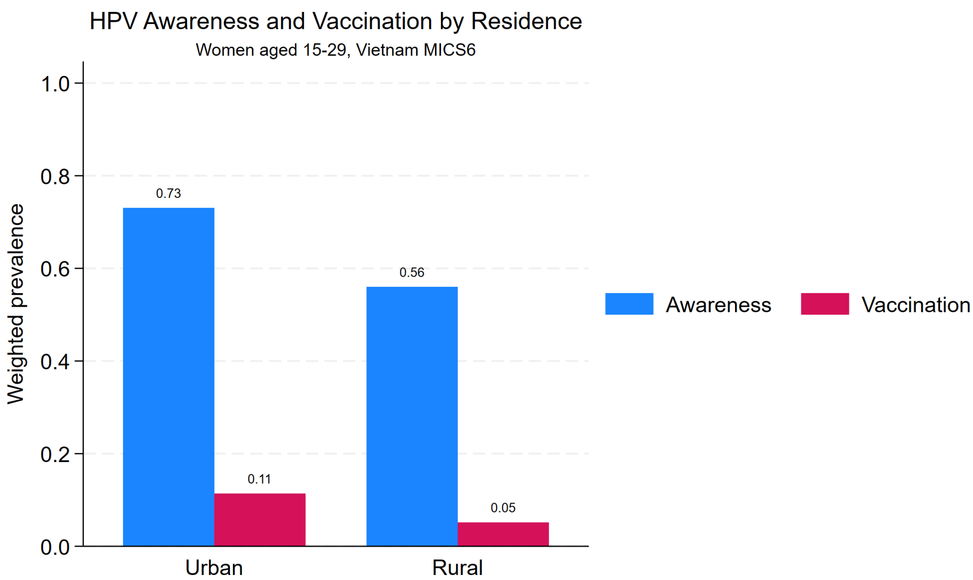

Figure 3.2. HPV vaccine awareness and vaccination prevalence by urban/rural residence.

Table 3.3. Awareness–vaccination gap among women aware of HPV vaccination

Variable	Category	Aware but unvaccinated % (95% CI)	p
Age	15–19	88.3 (83.9–91.7)	0.052

Variable	Category	Aware but unvaccinated % (95% CI)	p
	20–24	84.9 (80.9–88.2)	
	25–29	90.0 (87.4–92.1)	
Residence	Urban	84.4 (80.9–87.4)	0.002
	Rural	90.8 (88.2–92.9)	
Ethnicity	Kinh/Hoa	87.3 (85.0–89.3)	0.003
	Ethnic minority	95.5 (90.9–97.9)	
Health insurance	Insured	87.3 (85.0–89.2)	0.028
	Uninsured	93.1 (88.2–96.1)	
Education	Primary or less	96.6 (83.7–99.4)	< 0.001
	Lower secondary	95.1 (91.3–97.3)	
	Upper secondary+	86.3 (83.9–88.4)	
Wealth	Q1 poorest	97.9 (95.0–99.1)	< 0.001
	Q2	89.7 (85.0–93.0)	
	Q3	90.8 (86.8–93.6)	
	Q4	87.5 (83.3–90.8)	
	Q5 richest	80.2 (74.8–84.7)	
Mass media	No	93.3 (83.3–97.5)	0.195
	Yes	87.7 (85.6–89.6)	
Gender-equitable attitude	No	92.8 (87.2–96.1)	0.081
	Yes	87.9 (85.7–89.7)	
Ever had sex	No	84.0 (80.3–87.1)	< 0.001
	Yes	91.5 (89.4–93.1)	

Variable	**Category**	**Aware but unvaccinated % (95% CI)**	**p**
Note: Estimates use `svy, subpop(ccp5_bin)`. P-values are design-based Pearson chi-squared tests among women who had heard of HPV vaccination.			

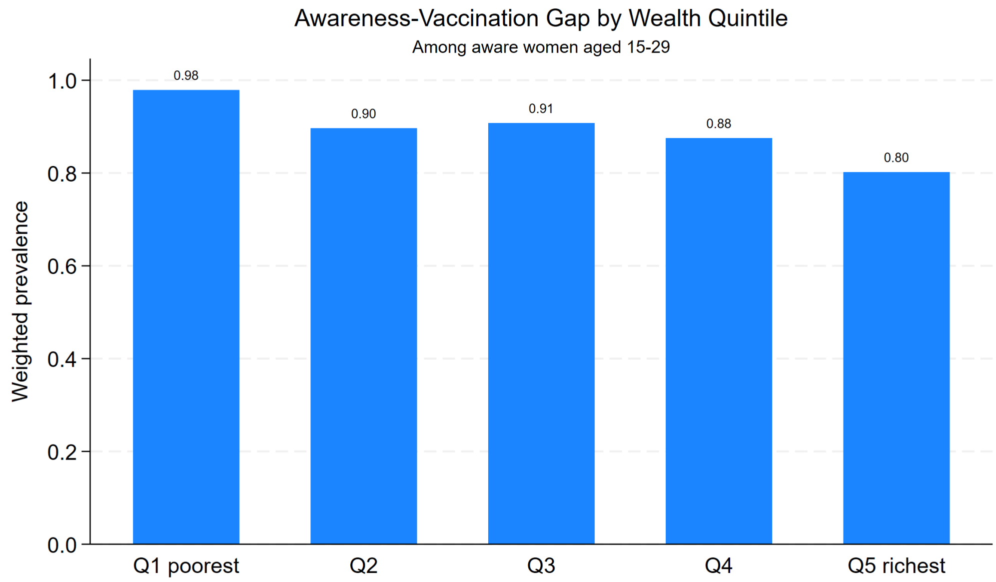

Figure 3.3. Awareness–vaccination gap by wealth quintile among women who had heard of HPV vaccination.

3.4 Socioeconomic inequality

Both awareness and vaccination were pro-rich. Vaccination showed stronger relative socioeconomic concentration than awareness based on the standard concentration index: 0.389 (95% CI: 0.296–0.482) for vaccination compared with 0.149 (95% CI: 0.131–0.166) for awareness. Erreygers-corrected indices should be interpreted as absolute inequality measures for bounded outcomes and are influenced by outcome prevalence; these were 0.371 (95% CI: 0.328–0.415) for awareness and 0.116 (95% CI: 0.089–0.144) for vaccination. The gap had a negative concentration index, showing that, among aware women, being unvaccinated was concentrated among poorer women.

Table 3.4. Concentration indices for HPV vaccine awareness, vaccination, and awareness–vaccination gap

Outcome	Index	N	Estimate (95% CI)	p
Awareness	Standard CI	4,102	0.149 (0.131–0.166)	< 0.001
Awareness	Erreygers CI	4,102	0.371 (0.328–0.415)	< 0.001
Vaccination	Standard CI	4,102	0.389 (0.296–0.482)	< 0.001
Vaccination	Erreygers CI	4,102	0.116 (0.089–0.144)	< 0.001
Gap	Standard CI	2,133	-0.035 (-0.047 to -0.023)	< 0.001
Gap	Erreygers CI	2,133	-0.123 (-0.164 to -0.082)	< 0.001

Note: Ranking variable was the continuous household wealth score. Estimates used the survey design. Positive values indicate pro-rich concentration; negative values indicate pro-poor concentration.

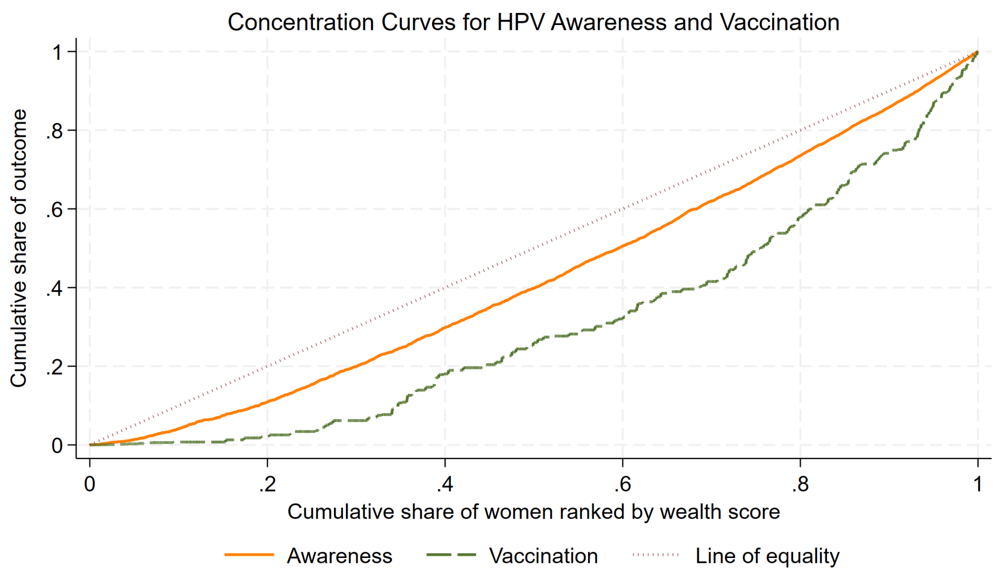

Figure 3.4. Concentration curves for HPV vaccine awareness and vaccination ranked by household wealth score.

3.5 Adjusted regression models

Table 3.5 presents survey-weighted Poisson regression models. All estimates are adjusted prevalence ratios. The awareness and vaccination models used the full analytic sample with complete model covariates (N = 3,899). The gap model used the aware subpopulation while preserving the survey design (N = 4,033 in the design sample; subpopulation observations were aware women with complete model covariates). P-values are model-based t-tests from survey-weighted Poisson regression with Taylor linearized survey variance.

Table 3.5. Adjusted prevalence ratios for HPV awareness, vaccination, and awareness–vaccination gap

Variable	**Awareness aPR (95% CI; p)**	**Vaccination aPR (95% CI; p)**	**Gap PR (95% CI; p)**
Rural	0.933 (0.867–1.005; 0.068)	0.780 (0.547–1.112; 0.169)	1.024 (0.976–1.075; 0.331)
Age 15–19	0.626 (0.561–0.699; < 0.001)	0.514 (0.305–0.865; 0.012)	1.053 (0.979–1.134; 0.166)
Age 20–24	0.918 (0.855–0.986; 0.019)	1.150 (0.796–1.660; 0.456)	0.975 (0.929–1.023; 0.299)
Formerly married/union	0.882 (0.730–1.065; 0.192)	1.482 (0.697–3.150; 0.306)	0.955 (0.856–1.066; 0.412)
Never married/union	1.050 (0.868–1.270; 0.615)	1.195 (0.578–2.470; 0.631)	0.973 (0.886–1.069; 0.568)
Primary or less	0.475 (0.330–0.683; < 0.001)	0.327 (0.060–1.770; 0.194)	1.008 (0.940–1.081; 0.816)
Lower secondary	0.836 (0.754–0.927; 0.001)	0.422 (0.227–0.785; 0.007)	1.041 (1.003–1.081; 0.036)
Uninsured	0.997 (0.898–1.108; 0.962)	0.676 (0.381–1.197; 0.179)	1.039 (0.991–1.090; 0.112)
Ethnic minority	0.860 (0.745–0.992; 0.038)	0.743 (0.307–1.801; 0.510)	1.010 (0.957–1.066; 0.716)
Q1 poorest	0.576 (0.480–0.691; < 0.001)	0.118 (0.039–0.358; < 0.001)	1.137 (1.055–1.225; 0.001)
Q2	0.802 (0.726–0.885; < 0.001)	0.585 (0.365–0.936; 0.026)	1.070 (0.992–1.154; 0.079)
Q3	0.812 (0.743–0.888; < 0.001)	0.437 (0.278–0.688; < 0.001)	1.104 (1.028–1.185; 0.007)
Q4	0.890 (0.829–0.956; 0.001)	0.631 (0.421–0.945; 0.026)	1.073 (0.994–1.158; 0.069)
Mass media exposure	1.147 (0.981–1.340; 0.084)	1.960 (0.763–5.034; 0.162)	0.962 (0.919–1.007; 0.097)
Gender-equitable attitude	1.234 (1.066–1.429; 0.005)	1.614 (0.861–3.025; 0.135)	0.978 (0.929–1.029; 0.391)
Ever had sex	1.024 (0.857–1.224; 0.793)	0.739 (0.352–1.550; 0.423)	1.050 (0.945–1.166; 0.361)
Reference categories: urban, age 25–29, currently married/in union, upper secondary or higher, insured, Kinh/Hoa, Q5 richest, no mass media exposure, no gender-equitable attitude proxy, and never had sexual intercourse. ICT was excluded from adjusted models because weighted prevalence was 97.9%.			

After adjustment, wealth remained the dominant predictor. Compared with the richest quintile, the poorest quintile had 42% lower awareness and 88% lower vaccination prevalence. Among aware women, the poorest quintile had a higher prevalence of remaining unvaccinated. Lower secondary education was associated with lower vaccination and a higher gap. The gender-equitable attitude proxy was independently associated with higher awareness but not with vaccination after adjustment.

3.6 Absolute differences and vaccination among aware women

Because vaccination was rare in the poorest quintile and the gap outcome was very common, absolute differences help interpret the relative models. In bivariate survey-weighted estimates among women who had heard of HPV vaccination, vaccination increased from 2.1% in the poorest quintile (95% CI: 0.9–5.0) to 19.8% in the richest quintile (95% CI: 15.3–25.2), with a design-based Pearson p-value < 0.001. This sensitivity analysis confirms that wealth differences persist even after conditioning on awareness.

Table 3.6 reports adjusted predicted probabilities from survey-weighted logistic models. Compared with the richest quintile, the poorest quintile had a 31.5 percentage-point lower adjusted probability of awareness, a 10.8 percentage-point lower adjusted probability of vaccination in the full analytic sample, and an 11.8 percentage-point lower adjusted probability of vaccination among aware women. The adjusted gap probability was 11.8 percentage points higher in the poorest quintile than in the richest quintile.

Table 3.6. Adjusted predicted probabilities by wealth quintile

Wealth	Aware % (95% CI)	Vaccinated % (95% CI)	Vaccinated if aware % (95% CI)	Gap % (95% CI)
Q1 poorest	48.0 (41.2–54.9)	1.5 (0.0–3.1)	3.1 (0.0–6.2)	96.9 (93.8–100.0)
Q2	61.1 (56.6–65.7)	7.1 (4.4–9.8)	11.1 (7.0–15.2)	88.9 (84.8–93.0)
Q3	61.8 (56.7–66.8)	5.3 (3.4–7.2)	8.3 (5.2–11.4)	91.7 (88.6–94.8)
Q4	68.6 (63.7–73.6)	7.7 (5.2–10.1)	10.4 (7.0–13.8)	89.6 (86.2–93.0)
Q5 richest	79.5 (74.9–84.1)	12.3 (8.8–15.7)	15.0 (10.7–19.2)	85.0 (80.8–89.3)

Note: Estimates are adjusted predicted probabilities from survey-weighted logistic regression models with the same covariates as the primary Poisson models. Gap and vaccination-among-aware models used the aware subpopulation while preserving the survey design. Confidence interval bounds for probabilities are shown within the 0–100% scale.

3.7 Erreygers decomposition

Decomposition results are summarized in Table 3.7. Percentages indicate each determinant's contribution to the total Erreygers concentration index. Positive percentage values contribute in the direction of the total index; negative values offset it.

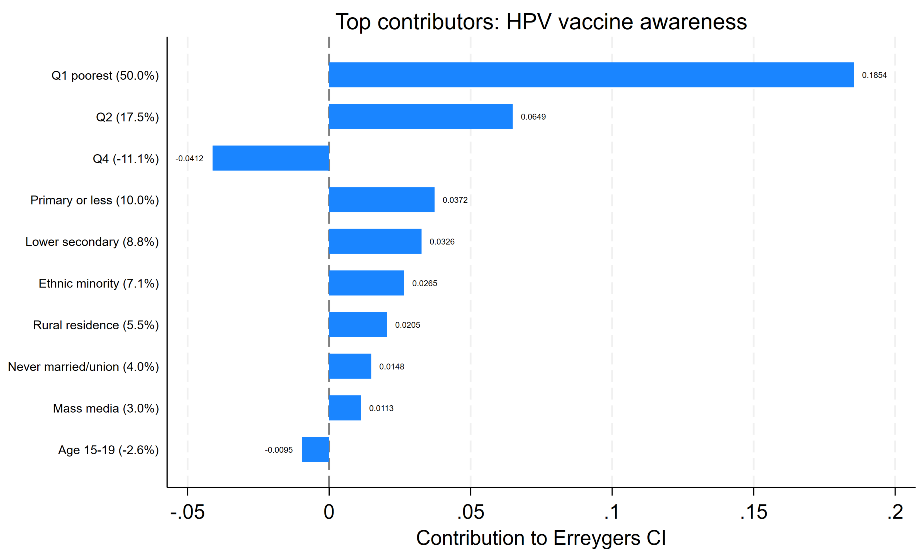

Figure 3.5. Top contributors to socioeconomic inequality in HPV vaccine awareness.

Table 3.7. Main contributors to Erreygers inequality decomposition

Outcome	Contributor	Contribution	% of total
Awareness	Q1 poorest	0.1854	50.0
	Q2	0.0649	17.5
	Primary or less	0.0372	10.0
	Lower secondary	0.0326	8.8
	Ethnic minority	0.0265	7.1
	Rural residence	0.0205	5.5
	Residual	0.0285	7.7
Vaccination	Q1 poorest	0.0770	66.1
	Lower secondary	0.0164	14.1
	Q2	0.0154	13.2
	Q4	-0.0132	-11.3
	Primary or less	0.0093	8.0
	Rural residence	0.0079	6.8
	Residual	-0.0203	-17.4
Gap	Q1 poorest	-0.0546	44.4
	Lower secondary	-0.0200	16.2
	Q2	-0.0188	15.2
	Q4	0.0121	-9.9
	Q3	-0.0109	8.8
	Rural residence	-0.0076	6.2
	Residual	0.0007	-0.6

Note: Decomposition used dummy-coded determinants, probit average marginal effects, and survey-weighted standard concentration indices for each determinant; contributions sum to the Erreygers concentration index of each outcome plus a residual. Reference groups match the adjusted models. Bootstrap confidence intervals were not estimated; contributions should be interpreted as explanatory accounting rather than causal attribution. Full output is available in the generated Stata CSV file.

The decomposition indicates that wealth was central for both awareness and vaccination inequality. For awareness, the poorest wealth quintile, Q2, primary or lower secondary education, ethnic minority status, and rural residence were the largest contributors. For vaccination, the poorest quintile alone accounted for approximately two-thirds of the observed pro-rich inequality, followed by lower secondary education and Q2; Q4 had an offsetting negative contribution because it was less disadvantaged than the poorest groups but still below the richest reference group. For the gap, negative contributions from poorer wealth groups, lower secondary education, and rural residence were consistent with the pro-poor concentration of remaining unvaccinated among women who were already aware.

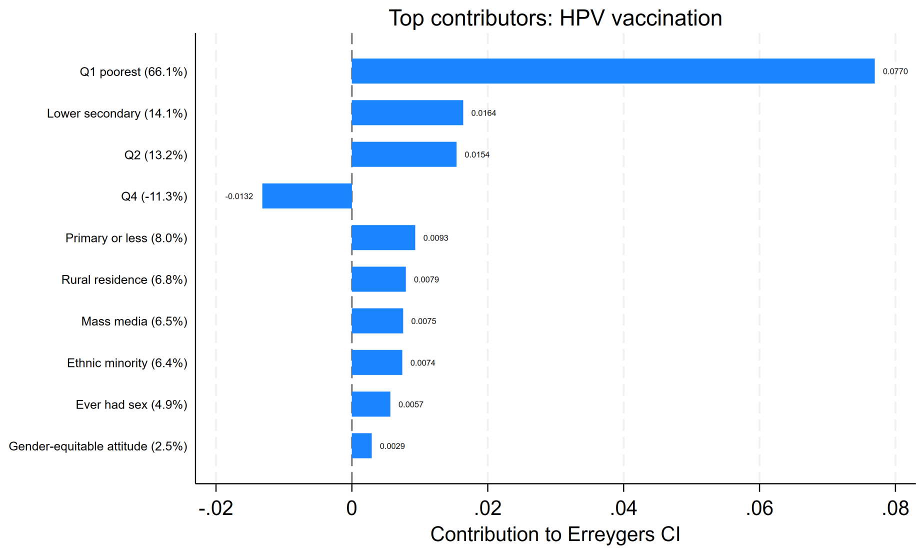

Figure 3.6. Top contributors to socioeconomic inequality in HPV vaccination.

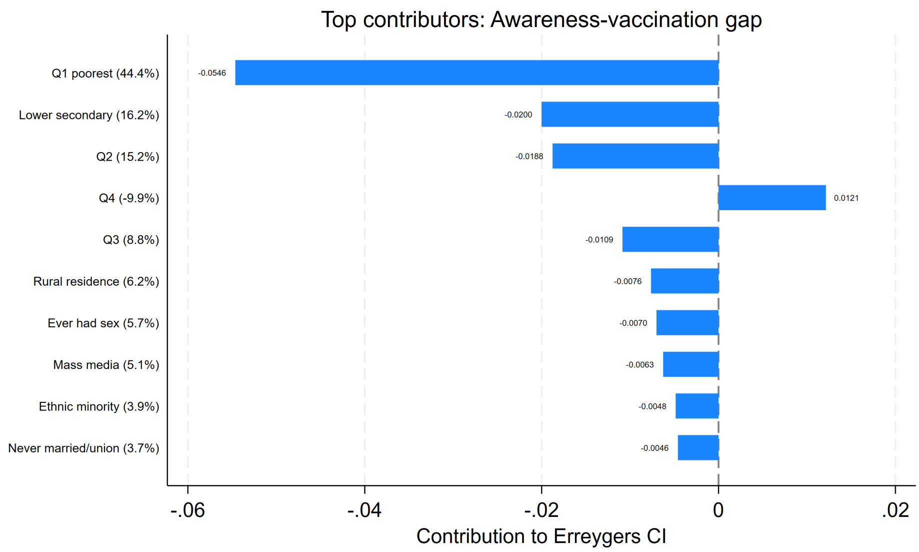

Figure 3.7. Top contributors to socioeconomic inequality in the awareness–vaccination gap.

CHAPTER 4

DISCUSSION

4.1 Principal findings

This thesis found a clear mismatch between HPV vaccine awareness and HPV vaccination among Vietnamese women aged 15–29 years. Nearly two-thirds of women had heard of HPV vaccination, but fewer than one in thirteen had been vaccinated. Among women who were already aware, only 12.0% had received the vaccine. This pattern demonstrates that information alone is not sufficient to generate uptake.

The second major finding is the strong socioeconomic gradient. Awareness was higher among women in richer households, urban residents, Kinh/Hoa women, and women with higher education. Vaccination showed an even steeper relative gradient: 16.6% of women in the richest quintile reported vaccination compared with only 0.7% in the poorest quintile. The standard concentration index confirmed that vaccination had stronger relative socioeconomic concentration than awareness. The Erreygers indices should be interpreted more cautiously as absolute inequality measures for bounded outcomes because they are influenced by outcome prevalence.

The third major finding is that the awareness–vaccination gap was concentrated among poorer aware women. This distinction matters. A woman who has never heard of the vaccine requires information and communication. A woman who has heard of the vaccine but remains unvaccinated may face financial, geographic, service delivery, social, or prioritization barriers. The gap outcome therefore points toward access and affordability, not only health education.

4.2 Interpretation of associated factors

Wealth was the most consistent marker across the pathway. In adjusted models, the poorest women had substantially lower awareness and vaccination than the richest women. They were also more likely to remain unvaccinated despite awareness. The decomposition results reinforced this conclusion: the poorest quintile was the largest contributor to both awareness and vaccination inequality. Because MICS6 predates the planned public financing of HPV vaccination, the observed socioeconomic gradient should be interpreted as a baseline measure of inequity under a predominantly opportunistic or privately financed access context. This is consistent with the access framework of Penchansky and Thomas, in which affordability, availability, and accessibility all shape use of health services [23].

Education also played an important role. Women with primary education or less had markedly lower awareness in the adjusted model. Lower secondary education was associated with lower vaccination and a higher awareness–vaccination gap. Education may operate through health literacy, confidence in preventive care, ability to interpret vaccine information, and capacity to navigate services [21].

Age patterns suggest possible missed catch-up opportunities. Vaccination prevalence was lowest among women aged 15–19 years and highest among those aged 20–24 years, which may reflect delayed vaccination after adolescence, differences in family decision-making, private-market affordability, or contact with reproductive health services. Because MICS6 is cross-sectional and does not record age at vaccination, these mechanisms cannot be separated. Programmatically, however, the pattern supports monitoring not only routine target-age vaccination but also older adolescents and young adults who may be missed during the transition to public financing.

Rural residence and ethnic minority status were strongly associated with lower awareness and vaccination in bivariate analysis. Ethnic minority status should be interpreted as a marker of structural marginalization, geographic access, language barriers, service availability, and historical inequities rather than as an intrinsic cause of lower vaccination. After adjustment, ethnic minority status remained associated with lower awareness but not vaccination, suggesting that some of the vaccination difference may be mediated through wealth, education, residence, and other structural variables. Nevertheless, ethnic minority women had very low vaccination prevalence in descriptive analysis, and this remains programmatically important.

Media and ICT exposure showed different roles. ICT exposure was almost universal and therefore not suitable for adjusted regression. Mass media exposure was strongly associated with awareness in bivariate analysis, but its adjusted association was weaker. In the decomposition, media exposure contributed meaningfully to inequality, suggesting that unequal distribution of information channels still matters even when other factors are controlled.

The gender-equitable attitude measure was based on rejecting all five wife-beating justifications. It was associated with higher awareness after adjustment. This variable should not be interpreted as direct decision-making autonomy; rather, it may capture broader gender norms, empowerment, and openness to preventive health information.

4.3 Comparison with previous evidence

The findings align with international evidence that HPV vaccination coverage is shaped by financing, delivery strategy, and social determinants [16, 17]. Countries with organized and publicly financed programs have achieved larger and more equitable population impact, while reliance on individual payment and opportunistic uptake can produce inequitable coverage. WHO's 2024 product-choice guidance and single-dose updates are relevant for Viet Nam because simplified schedules can reduce procurement, service delivery, and follow-up barriers that often disadvantage poorer and remote communities [12, 15]. The results also support previous methodological work showing that odds ratios can exaggerate associations for common outcomes; therefore, prevalence ratios are preferable for cross-sectional analyses of common outcomes such as awareness [28, 29].

The magnitude of the awareness–vaccination gap in this thesis highlights a specific implementation problem. Raising awareness remains necessary, especially among poor, rural, less educated, and ethnic minority women. However, awareness campaigns alone will not close the vaccination gap unless they are combined with affordable and accessible vaccination services.

4.4 Policy implications

The results support an equity-oriented HPV vaccination strategy with four components. Resolution 104/NQ-CP places HPV vaccination on Viet Nam's pathway for inclusion in the Expanded Programme on Immunization from 2026 [5, 6]. The implication is not simply that public financing is needed, but that implementation should be designed so that public financing does not reproduce the socioeconomic gradient observed in the MICS6 baseline.

First, financing matters. Public financing or strong subsidy mechanisms are likely to be necessary if Viet Nam aims to prevent HPV vaccination from functioning primarily as a private-market service accessible to better-off households.

Second, delivery should prioritize geographic access. Rural and remote areas require delivery channels that reduce travel time and indirect costs. School-based, community-based, or primary health care-based delivery can be considered depending on local implementation capacity.

Third, communication should be targeted rather than only universal. Women in poorer households, ethnic minority communities, and lower education groups had lower awareness. Communication materials should be culturally appropriate, understandable, and available through trusted community and health system channels.

Fourth, monitoring should include equity indicators. National or provincial reporting should not only track overall doses or coverage, but also coverage by wealth, ethnicity, education, residence, and school attendance. Equity monitoring is essential for cervical cancer elimination goals [3]. Particular attention is needed for poor households, rural communes, ethnic minority communities, out-of-school girls, and young women who are older than the initial school-based target age.

4.5 Strengths and limitations

This study used nationally representative MICS6 data and accounted for the complex survey design in all main analyses. The analysis also moved beyond simple prevalence estimation by using concentration indices and decomposition to quantify socioeconomic inequality.

Several limitations should be considered. First, the cross-sectional design does not permit causal inference; terms such as associated factors, determinants, and contributors in this thesis refer to statistical and decomposition patterns, not proven causal effects. Second, HPV vaccination status was self-reported and may be affected by recall or reporting error. Third, because HPV vaccination was self-reported and detailed information on dose number, age at vaccination, vaccine type, price paid, provider type, and reasons for non-vaccination was unavailable, the analysis could not distinguish incomplete vaccination, delayed vaccination, affordability barriers, service availability constraints, and vaccine hesitancy. Fourth, the full-sample vaccination outcome coded women who had never heard of HPV vaccination as unvaccinated, consistent with the skip pattern; this is appropriate for estimating reported vaccination in the full population, but it combines awareness and uptake. The vaccination-among-aware sensitivity analysis was therefore added to separate uptake conditional on awareness. Fifth, sexual experience was self-reported in a face-to-face survey and may be affected by underreporting, particularly among unmarried young women in settings where premarital sex is socially sensitive. Sixth, ICT exposure was nearly universal and could not be reliably included in adjusted models. Seventh, decomposition results depend on model specification and should be interpreted as explanatory accounting rather than causal attribution.

CONCLUSION AND RECOMMENDATIONS

Conclusion

This secondary analysis of Viet Nam MICS6 found that HPV vaccine awareness among women aged 15–29 years was moderate, but vaccination uptake was low. Overall, 62.4% of women had heard of HPV vaccination, while only 7.5% had received HPV vaccine. Among women who were aware of the vaccine, 88.0% remained unvaccinated.

Socioeconomic inequality was substantial. Awareness and vaccination were concentrated among wealthier women; vaccination showed stronger relative socioeconomic concentration than awareness, while Erreygers indices described absolute inequality for these bounded outcomes. The awareness–vaccination gap was concentrated among poorer aware women, indicating that barriers beyond information prevent vaccine uptake.

Adjusted models and decomposition consistently identified household wealth, education, residence, mass media exposure, ethnicity, and gender-equitable attitudes as important components of the inequality pattern. The poorest wealth quintile was the most important contributor to pro-rich inequality in vaccination.

Recommendations

1. **Use public financing to reduce, not reproduce, inequity.** As HPV vaccination is incorporated into public immunization financing from 2026, implementation should prioritize poor households, rural areas, ethnic minority communities, and provinces or communes with weaker access.

2. **Target the awareness–vaccination gap.** Programs should not stop at information campaigns. Women who already know about the vaccine need accessible, low-cost, and convenient vaccination opportunities.

3. **Strengthen school-based and community delivery.** School-based delivery is appropriate for the main target age group, but catch-up strategies should be considered for girls and young women missed by school-based delivery, including out-of-school adolescents and older cohorts depending on available budget.

4. **Develop equity-focused communication.** Communication should be tailored for lower education groups, ethnic minority communities, and rural women, using clear language, appropriate local languages, and trusted community and health system channels.

5. **Monitor equity indicators.** Future HPV vaccine monitoring should report not only national coverage but also equity-stratified coverage by wealth quintile, residence, ethnicity, education, and school attendance, with special attention to girls and young women missed by school-based delivery.

6. **Improve future data collection.** Future surveys should capture dose number, age at vaccination, vaccine type, payment source, service location, and reasons for non-vaccination to better distinguish hesitancy, affordability, and service access barriers.

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CHAPTER A

APPENDICES

A.1 Generated Stata outputs

The final analysis pipeline saved outputs under `HPV thesis/Output`. The key files used for this thesis were:

- `tables/table1_descriptive.csv`
  Weighted descriptive percentages and 95% confidence intervals.
- `tables/table2_awareness_bivariate.csv`
  Awareness prevalence by explanatory variables, 95% confidence intervals, and design-based Pearson p-values.
- `tables/table3_vaccination_bivariate.csv`
  Vaccination prevalence by explanatory variables, 95% confidence intervals, and design-based Pearson p-values.
- `tables/table3b_gap_bivariate.csv`
  Awareness–vaccination gap among aware women.
- `tables/table4_concentration_indices.csv`
  Standard and Erreygers concentration indices.
- `tables/table5_models.csv`
  Adjusted Poisson models and logistic sensitivity models.
- `tables/table6_decomposition.csv`
  Erreygers decomposition results.
- `tables/table7_vaccination_aware_sensitivity.csv`
  Vaccination prevalence among women who had heard of HPV vaccination.

- `tables/table8_adjusted_predicted_by_wealth.csv`
  Adjusted predicted probabilities and absolute differences by wealth quintile.
- `figures/*.png`
  Figures copied into the LaTeX figure directory.

A.2 Logistic regression sensitivity analysis

Table A.1 reports odds ratios from survey-weighted logistic regression. These models were used as sensitivity analyses only; prevalence ratios from modified Poisson models were used for primary interpretation. The generated Stata output also includes a separate logistic model for vaccination among women who had heard of HPV vaccination.

Table A.1. Survey-weighted logistic regression sensitivity analysis

Variable	**Awareness OR (95% CI)**	**Vaccination OR (95% CI)**	**Gap OR (95% CI)**
Rural	0.776 (0.589–1.022)	0.754 (0.505–1.127)	1.199 (0.803–1.791)
Age 15–19	0.230 (0.160–0.332)	0.467 (0.259–0.841)	1.380 (0.769–2.478)
Age 20–24	0.715 (0.550–0.930)	1.178 (0.771–1.801)	0.800 (0.519–1.234)
Primary or less	0.249 (0.144–0.430)	0.314 (0.056–1.760)	1.256 (0.237–6.647)
Lower secondary	0.588 (0.448–0.771)	0.403 (0.211–0.771)	2.064 (1.050–4.057)
Ethnic minority	0.709 (0.521–0.965)	0.729 (0.284–1.870)	1.267 (0.500–3.214)

Variable	Awareness OR (95% CI)	Vaccination OR (95% CI)	Gap OR (95% CI)
Q1 poorest	0.198 (0.123–0.317)	0.104 (0.033–0.326)	5.676 (1.826–17.648)
Q2	0.361 (0.245–0.533)	0.534 (0.313–0.911)	1.426 (0.827–2.460)
Q3	0.372 (0.255–0.544)	0.387 (0.232–0.646)	1.985 (1.166–3.378)
Q4	0.525 (0.365–0.757)	0.581 (0.362–0.933)	1.538 (0.941–2.514)
Mass media exposure	1.395 (0.992–1.962)	2.050 (0.763–5.508)	0.482 (0.177–1.312)
Gender-equitable attitude	1.737 (1.219–2.474)	1.691 (0.860–3.325)	0.773 (0.399–1.500)
Ever had sex	1.273 (0.546–2.969)	0.703 (0.296–1.669)	1.344 (0.580–3.118)

Models used the same covariates and survey design as the primary Poisson models. Reference groups match Table 3.5.

A.3 Reproducibility note

The main do-file, raw input, and output directory are:

```
HPV thesis/Code/HPV_thesis_merged.do
HPV thesis/Dataset/data_raw_mics6_inter_format.dta
HPV thesis/Output
```

The analysis writes all generated files to the output directory and does not modify the raw dataset.